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*Discover Learning*

# Data Science with Python Certification Course

Course Curriculum : Your 16 module Learning Plan

https://www.edureka.co/data-science-python-certification-course

## About Edureka

Edureka is a leading e-learning platform providing live instructor-led interactive online training. We cater to professionals and students across the globe in categories like Big Data & Hadoop, Business Analytics, NoSQL Databases, Java & Mobile Technologies, System Engineering, Project Management and Programming. We have an easy and affordable learning solution that is accessible to millions of learners. With our students spread across countries like the US, India, UK, Canada, Singapore, Australia, Middle East, Brazil and many others, we have built a community of over 1 million learners across the globe.

## About Course

Edureka's Data Science with Python Certification Course, accredited by NASSCOM, is designed to help you master essential Python programming concepts for Data Science. You'll gain expertise in data operations, file handling, object-oriented programming, and key Python libraries like Pandas, NumPy, and Matplotlib. Whether you're a beginner or a professional, this course provides a comprehensive introduction to Machine Learning, Recommendation Systems, and other critical Data Science techniques to jumpstart your career in the field.


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# Data Science with Python Course Curriculum

## Module 1: Introduction to Python for Data Science

Topics:

- Set operations
- List comprehensions
- Generator functions
- Using modules
- Regex patterns
- Special collections
- Map & filter
- Custom exceptions

Hands-On/Demo:

- Writing a “Hello World” script
- Manipulating lists and dictionaries
- Reading a CSV file

Skills:

- Python programming
- Data science environment setup

## Module 2: Working with Python Programming

Topics:

- OOP concepts


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- Class inheritance
- Context managers
- Unit testing
- API requests
- Code profiling
- Logging basics
- JSON handling
- Project packaging
- Type hints

Hands-On/Demo:

- Using regex for data cleaning
- Creating a generator
- Building a custom module

Skills:

- Intermediate Python techniques
- Efficient code structures

## Module 3: Advanced Python Programming for Data Science

Topics:

- NumPy arrays
- Vector operations
- Math functions
- Series handling
- DataFrames
- Dataset merging


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- Missing values
- Pivot tables
- Data summaries
- Memory tuning

Hands-On/Demo:

- Building a preprocessing class
- Fetching API data
- Writing a unit test

Skills:

- Advanced Python programming
- Modular code developmen

## Module 4: Data Analysis with NumPy and Pandas

Topics:

- NumPy arrays
- Vector operations
- Math functions
- Series handling
- DataFrames
- Dataset merging
- Missing values
- Pivot tables
- Data summaries
- Memory tuning


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Hands-On/Demo:

- NumPy array calculations
- Cleaning data with Pandas
- Creating a pivot table

Skills:

- Data manipulation
- Basic statistical analysis

## Module 5: Data Visualization and Preprocessing Techniques

Topics:

- Matplotlib Plotting
- Seaborn Visualization Styles
- Line and Bar Charts
- Histogram Analysis
- Web Scraping Basics
- Missing Data Treatment
- Feature Scaling Techniques
- Encoding Categorical Data
- Data Storytelling Approaches

Hands-On/Demo:

- Creating Seaborn plots
- Scraping website data
- Normalizing a dataset

Skills:


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- Data visualization
- Data preprocessing

## Module 6: Statistical Methods for Data Science

Topics:

- Descriptive stats
- Variance and standard deviation
- Probability
- Normal distribution
- Hypothesis testing: t-tests
- Correlation: Pearson coefficient
- Outlier detection: z-score
- Sampling: random sampling
- Statistical visualization
- P-values: significance

Hands-On/Demo:

- Conducting a t-test
- Visualizing correlations
- Detecting outliers

Skills:

- Statistical analysis
- Result interpretation


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## Module 7: Fundamentals of Machine Learning

Topics:

- CRISP-DM process
- ML categories
- Python for ML
- ML tools
- Data lifecycle
- Evaluation
- Feature basics
- AI ethics
- Industry insights

Hands-On/Demo:

- Setting up an ML project
- Exploring a dataset

Skills:

- ML workflows
- Industry trends

## Module 8: Supervised Learning – Regression Analysis

Topics:

- Linear regression
- Gradient descent
- Polynomial regression
- Ridge regression


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- Error metrics
- R-squared
- Cross-validation
- Residual analysis
- Feature selection
- Overfitting mitigation

Hands-On/Demo:

- Building a linear regression model
- Evaluating with RMSE

Skills:

- Regression modeling
- Model evaluation

## Module 9: Supervised Learning – Classification Fundamentals

Topics:

- Logistic regression
- Binary labels
- Decision trees
- Confusion matrix
- Precision & recall
- ROC curve
- Overfitting
- Feature ranking
- Model validation


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- Class imbalance

Hands-On/Demo:

- Logistic regression model
- Decision tree visualization

Skills:

- Binary classification
- Evaluation metrics

## Module 10: Supervised Learning – Advanced Classification

Topics:

- Random forests
- SVM
- XGBoost
- Grid search
- Random search
- SHAP values
- SMOTE
- Model stacking
- Association rules
- Recommendation engines
- Model evaluation

Hands-On/Demo:

- Random Forest model
- Using SHAP for insights


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- Building Apriori rules

Skills:

- Advanced classification
- Interpretability, recommendations

## Module 11: Unsupervised Learning and Clustering Techniques

Topics:

- K-Means clusters
- Elbow method
- Hierarchical clustering
- DBSCAN logic
- PCA reduction
- Anomaly detection
- Silhouette score
- Segmentation use
- Cluster visuals

Hands-On/Demo:

- K-Means clustering
- Applying PCA
- Detecting anomalies

Skills:

- Unsupervised learning
- Dimensionality reduction


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## Module 12: AutoML and No-Code Data Science Solutions

Topics:

- AutoML tools
- DataRobot
- KNIME workflows
- H2O.ai models
- Synthetic data
- Rapid prototyping
- AI fairness

Hands-On/Demo:

- DataRobot model building
- KNIME workflow creation
- Generating synthetic data

Skills:

- AutoML prototyping
- No Code workflows

## Module 13: Reinforcement Learning Essentials (Self-paced)

Topics:

- Agent-Environment Interaction
- OpenAI Gym Setup
- Markov Decision Process
- Q-Learning Fundamentals
- Exploration-Exploitation Tradeoff


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- Epsilon-Greedy Strategy
- Reward Shaping Concepts
- Reinforcement Learning Applications
- Q-Table Implementation
- Reinforcement Learning Limitations

Hands-On/Demo:

- Q-Learning in a game
- OpenAI Gym experiment

Skills:

- RL algorithms
- Practical applications

## Module 14: Time Series Analysis and Forecasting Methods (Self-paced)

Topics:

- Time Series Components
- Stationarity Testing (ADF)
- ARIMA Model Parameters
- Forecasting with Prophet
- Forecast Error Metrics
- Backtesting Techniques
- Trend Visualization Methods
- Confidence Interval Analysis
- External Variable Integration
- Model Selection Strategies


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Hands-On/Demo:

- ARIMA model
- Prophet forecasting
- Visualizing trends

Skills:

- Time series analysis
- Forecasting

## Module 15: Machine Learning on Cloud Platforms (Self-paced)

Topics:

- Cloud ML Introduction
- AWS SageMaker
- Google Cloud AI
- Azure ML
- Cloud storage
- Serverless ML
- Model deployment
- Scalability

Hands-On/Demo:

- Training a model in SageMaker
- Deploying with Google Cloud AI
- Using S3 for data storage

Skills:

- Cloud-based ML


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- Scalable model deployment

## Module 16: MLOps Fundamentals (Self-paced)

Topics:

- MLOps Introduction
- CI/CD for ML
- Flask API Deployment
- MLflow Model Tracking
- Docker Containerization
- Model Drift Monitoring
- Model Lifecycle Management

Hands-On/Demo:

- Deploying with Flask
- MLflow pipeline setup

Skills:

- MLOps practices


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